Overview: Modern information systems are essential to organizations, enabling communication, data processing, and decision-making. However, as systems become more interconnected (e.g., cloud computing, mobile platforms, IoT), they are increasingly exposed to cyber threats.

This module introduces the structure of modern information systems and emphasizes the importance of integrating **security principles** to ensure confidentiality, integrity, and availability of data.

Intended Learning Outcomes

By the end of this module, students should be able to:

- Describe the components of modern information systems
- Explain the role of security in system design and operation
- Identify common threats affecting modern systems
- Apply basic security principles to protect information asset

What is a Modern Information System?

A modern information system is an integrated set of components used to collect, process, store, and distribute information.

Key Components:

- Hardware – physical devices (servers, computers, mobile devices)
- Software – applications and operating systems
- Data – raw facts processed into meaningful information
- People – users and IT professionals
- Processes – procedures and workflows

Characteristics of Modern Systems

- **Cloud-based Infrastructure (e.g., SaaS, IaaS)**
  - Refers to the delivery of computing services (such as storage, servers, databases, networking, and software) over the internet instead of local machines.
    - SaaS (Software as a Service): Users access applications via a web browser (e.g., Google Docs).
    - IaaS (Infrastructure as a Service): Provides virtualized computing resources like servers and storage (e.g., AWS, Microsoft Azure).
  - Enables scalability, cost-efficiency, and remote accessibility.
- **Mobile and Remote Access**
  - Allows users to access systems and data from different locations using mobile devices (smartphones, tablets, laptops) through the internet.
  - Supports work-from-home setups and global collaboration but increases security risks such as unauthorized access and insecure networks.
- **Interconnected Networks (Internet, IoT)**
  - Modern systems are connected through networks, enabling communication between devices and systems worldwide.
    - Internet: Global network connecting computers and servers
    - IoT (Internet of Things): Network of physical devices (e.g., smart home devices, sensors) connected to the internet
  - Enhances automation and data sharing but expands the attack surface for cyber threats.
- **Real-time Data Processing**
  - The ability of systems to process and analyze data immediately as it is generated, without delay.
  - Used in applications like online banking, traffic monitoring, and e-commerce transactions, where instant decision-making is critical.

Why Security is Critical?

Modern systems face various risks due to connectivity and data exposure.

Core Security Goals (CIA Triad):

- Confidentiality – Protect data from unauthorized access
- Integrity – Ensure data is accurate and unaltered
- Availability – Ensure systems and data are accessible when needed

Common Threats to Information Systems

- **Malware**
  - Malware refers to malicious software designed to harm, disrupt, or gain unauthorized access to computer systems and data.
  - Common types:
    - Viruses – spread by attaching to files
    - Ransomware – locks files and demands payment
    - Spyware – secretly collects user information
- **Phishing and social engineering**
  - Phishing and social engineering involve tricking people into revealing sensitive information or performing unsafe actions.
  - **Common methods:**
    - Fake emails or websites
    - Impersonation of trusted organizations
    - SMS or phone scams
- **Unauthorized access / hacking**
  - Unauthorized access (hacking) occurs when someone gains entry to a system or network without permission.
  - Common methods:
    - Password cracking
    - Exploiting software vulnerabilities

    - Brute-force attacks
    - Data breaches
    - Insider threats
- **Data Breach**
  - A data breach is the exposure or theft of confidential information without authorization.
  - Common causes:
    - Weak security controls
    - Misconfigured databases (especially in cloud systems)
    - Cyberattacks
- **Insider Threats**
  - Insider threats come from employees or trusted individuals who misuse their access to systems or data.
  - Types:
    - Malicious insiders (intentional harm)
    - Negligent insiders (careless mistakes)

Modern Systems Security Risks

- **Cloud-based Infrastructure**
  - Data breaches due to misconfigured cloud storage
  - Unauthorized access from weak authentication
  - Data loss or service outages
  - Shared environment vulnerabilities (multi-tenancy risks)
- **Mobile and Remote Access**
  - Use of insecure public Wi-Fi
  - Device theft or loss
  - Unauthorized access from unmanaged devices
  - Malware infections on mobile endpoints
- **Interconnected Networks (Internet, IoT)**
  - Increased attack surface for cyberattacks
  - Weak security in IoT devices
  - Distributed Denial-of-Service (DDoS) attacks
  - Unauthorized device access
- **Real-time Data Processing**
  - Data interception during transmission
  - Integrity issues due to rapid processing
  - System overload or denial-of-service attacks
  - Inadequate validation of real-time inputs

Basic Security Mechanisms

- **Authentication**
  - Authentication is the process of verifying the identity of a user or system to ensure they are who they claim to be.
  - **Examples:** Passwords and PINs, Fingerprint or facial recognition (biometrics), One-time passwords (OTP)
  - **Purpose:** To prevent unauthorized users from accessing systems.
- **Authorization**
  - Authorization determines what an authenticated user is allowed to do or access within a system.
  - **Examples:** Admin vs. regular user permissions, and Read, write, or execute access to files , etc.
  - **Purpose:** To ensure users only access information relevant to their role.
- **Firewalls and intrusion detection systems**
  - Firewalls: Security systems that monitor and control incoming and outgoing network traffic.
  - Intrusion Detection Systems (IDS): Tools that detect suspicious or malicious activities in a network.
  - **Examples:** Blocking unauthorized access attempts, Alerting administrators of unusual behavior, etc.
  - **Purpose:** To protect networks from external and internal threats.
- **Backup and recovery systems**
  - Backup and recovery systems involve creating copies of data and restoring them in case of data loss or system failure.
  - **Examples:** Cloud backups (Google Drive, Dropbox), External hard drive backups, Disaster recovery plans
  - **Purpose:** To ensure data can be restored after cyberattacks, hardware failure, or accidental deletion.
- **Encryption**
  - Encryption is the process of converting data into a coded format so that only authorized parties can read it.
  - **Examples:** HTTPS (secure websites), Encrypted messaging apps, File encryption, etc.
  - **Purpose:** To protect sensitive data from being read if intercepted.

SUMMARY / KEY TAKEAWAYS

- Modern information systems integrate technology, people, and processes
- Increased connectivity introduces greater security risks
- The CIA triad is fundamental to protecting information
- Security must be built into systems, not added later
- Understanding threats helps in designing effective defenses

Overview: This module introduces the principles and applications of modern cryptography used to secure information systems. Students will explore encryption techniques, key algorithms such as RSA, Elliptic Curve Cryptography (ECC), and AES, and analyze cryptographic protocols and their vulnerabilities.

Intended Learning Outcomes

By the end of this module, students should be able to:

- Explain fundamental cryptographic concepts
- Differentiate between symmetric and asymmetric encryption
- Describe how RSA, ECC, and AES work
- Analyze cryptographic protocols for security weaknesses
- Implement basic cryptographic algorithms using code or pseudocode

What is Cryptography?

Cryptography is the practice of securing communication by transforming information into a format that prevents unauthorized access. It uses mathematical techniques (like encryption algorithms and keys) to hide the original meaning of data during storage or transmission.

Key Goals:

- Confidentiality – Ensures that only authorized users can access the information and prevents unauthorized disclosure
- Integrity – Ensures that data has not been altered during transmission or storage and detects any unauthorized changes
- Authentication – Confirms that users or systems are who they claim to be
- Non-repudiation – Ensures that a sender cannot deny sending a message or performing an action

These four goals work together to make digital communication secure, trustworthy, and reliable.

Types of Encryptions

1. **Symmetric encryption** is a method of securing data where the same single key is used for both encryption and decryption.
   - **Key Characteristics:**
     - *Single Shared Key*
       - The sender and receiver use the same secret key
       - This key must be kept secure and shared safely
     - *Fast and Efficient*
       - Much quicker than asymmetric encryption
       - Ideal for encrypting large amounts of data (e.g., files, databases, streaming)
   - **Analogy:** Like a shared locker key—both sender and receiver use the same key.
   - **Example Algorithm:** AES
   - **Uses:** File encryption (e.g., hard drives, USBs), Secure communication after connection is established (e.g., HTTPS sessions), Data storage systems etc.
   - **Limitation:**
     - Key distribution problem
     - Sharing the secret key securely can be difficult

2. Asymmetric encryption is a method of securing data that uses two different but related keys: a public key and a private key.
   - **Key Characteristics**
     - Two-Key System (Key Pair)
       - Public key → shared openly, used to encrypt data
       - Private key → kept secret, used to decrypt data
     - More Secure Key Distribution
       - No need to share the private key
       - Reduces the risk compared to symmetric encryption
     - Slower Performance
       - More computationally intensive
       - Less efficient for large amounts of data
   - **Example Algorithms:** RSA, ECC
   - **Analogy:** Like a **mailbox**—anyone can drop a letter (public key), but only the owner can open it (private key).
   - **Uses:** Secure key exchange (e.g., establishing HTTPS connections), Digital signatures, Email encryption
   - **Limitation:** Slower than symmetric encryption and Not ideal for encrypting large data directly

Cryptographic Algorithms

1. **RSA (Rivest–Shamir–Adleman)**
   - RSA is a widely used asymmetric encryption algorithm that secures data by using a pair of keys: a public key for encryption and a private key for decryption.
   - RSA is based on the mathematical principle that: "*Multiplying large prime numbers is easy, but factoring their product is extremely difficult.* "
   - **Key Features:**

  - Uses two keys (public and private)
  - Enables secure communication over insecure networks
  - Supports both encryption and digital signatures
- **Use Cases**
  - Secure Key Exchange - Used in HTTPS to safely share symmetric keys
  - Digital Signatures - Verifies authenticity and ensures non-repudiation
  - Secure Email Communication - Encrypting sensitive messages
- **Key Generation (Simplified):**
  1. Choose two large primes (p, q)
  2. Compute n = p × q
  3. Compute Euler's Totient ϕ(n) = (p−1)(q−1)
  4. Choose public key (e)
  5. Compute private key (d)
- **Example (Using Small Numbers)**
  - Step 1: Choose Two Prime Numbers: $\boldsymbol{p = 3, q = 11}$
  - Step 2: Compute n: $\boldsymbol{n = p \times q = 3 \times 11 = 33}$
  - Step 3: Compute Euler's Totient ϕ(n):
    $\boldsymbol{\phi(n) = (p-1)(q-1) = (3-1)(11-1) = 2 \times 10 = 20}$
  - Step 4: Choose Public Key (e)
    - Choose $e = 3$; It must be relatively prime to 20
  - Step 5: Compute Private Key d
    - $(d \times e) mod 20 = 1$
    - $7 \times 3 = 21 \equiv 1$
  - **Final Keys:**
    - Public Key: (e = 3, n = 33)
    - Private Key: (d = 7, n = 33)
  - **Encryption:**
    Message: $M = 5$
    Public Key: (e = 3, n = 33)
    Encryption formula: $C = M^e \bmod n$
    $5^3 = 125$; $125 \bmod 33$
    $C = 5^3 \bmod 33 = 26$
    Ciphertext = 26

    Decryption
    Message: $C = 26$
    Private Key: (d = 7, n = 33)
    Decryption formula: $M = C^d \ mod \ n$
    $= 26^7 mod \ 33$
    $M = 5$

Note: Real RSA uses very large prime numbers (hundreds of digits)